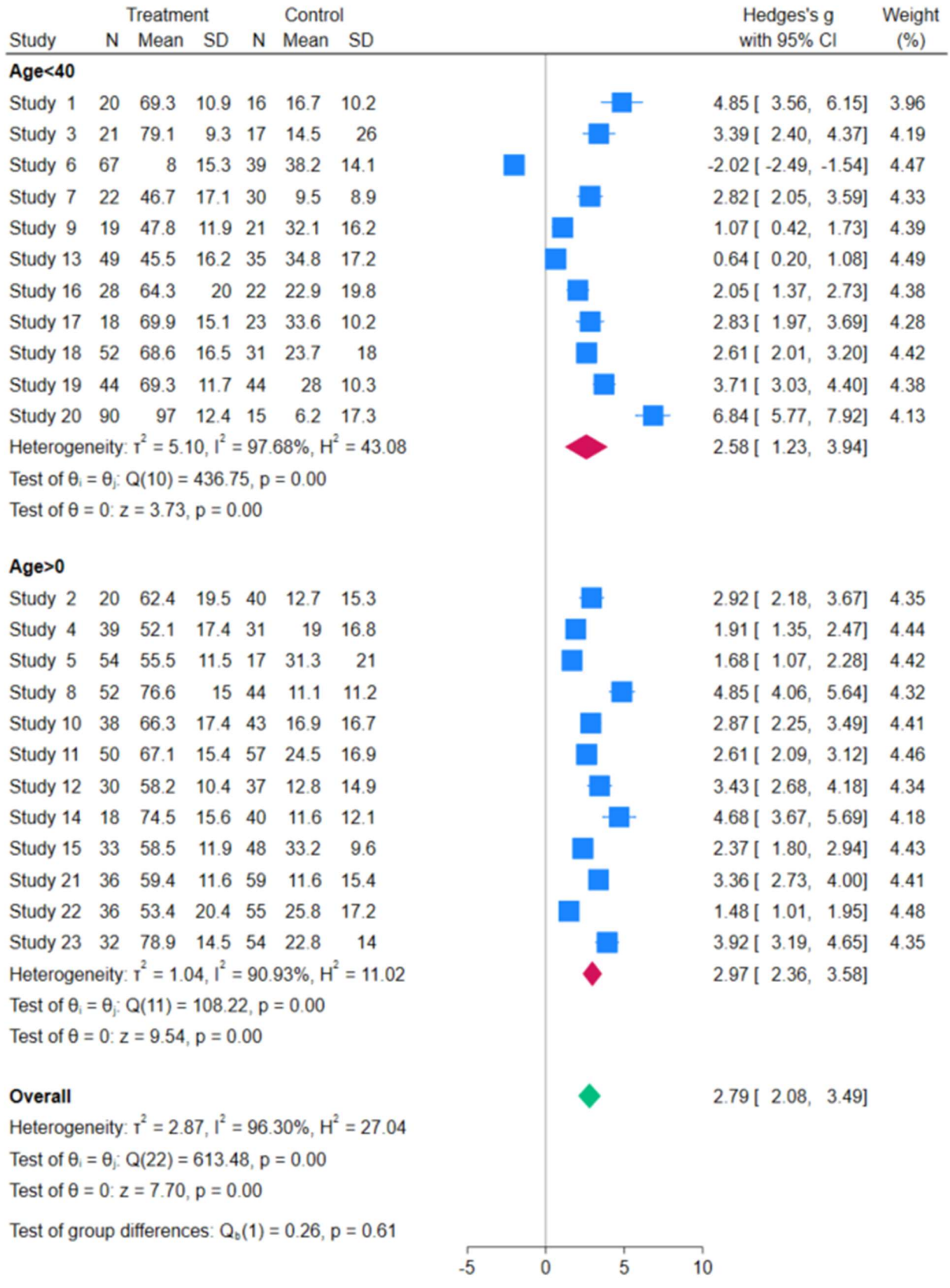
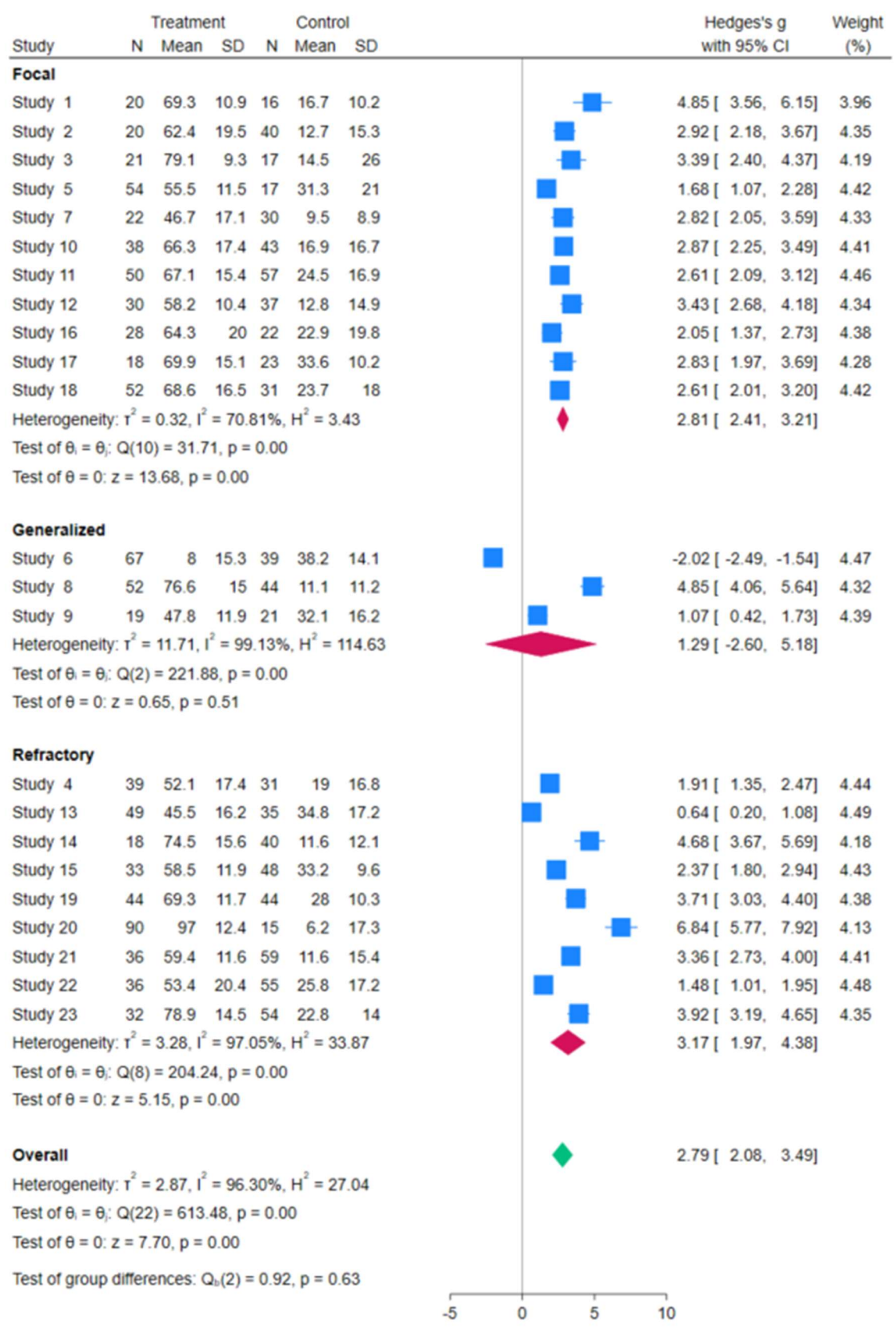
Name:Hamid Reza Ghahramani

Task 5:

Please send screenshots of each step of your task.

1. Conduct subgroup analysis and meta-regression based on the variables: epilepsy type and age (<=40&>40), and interpret your findings.


Treatment
Control
Study N Mean SD N Mean SD
Hedges's g with 95% CI
Weight (%)
Age<40
Study 1 20 69.3 10.9 16 16.7 10.2 4.85 [3.56, 6.15] 3.96
Study 3 21 79.1 9.3 17 14.5 26 3.39 [2.40, 4.37] 4.19
Study 6 67 8 15.3 39 38.2 14.1 -2.02 [-2.49, -1.54] 4.47
Study 7 22 46.7 17.1 30 9.5 8.9 2.82 [2.05, 3.59] 4.33
Study 9 19 47.8 11.9 21 32.1 16.2 1.07 [0.42, 1.73] 4.39
Study 13 49 45.5 16.2 35 34.8 17.2 0.64 [0.20, 1.08] 4.49
Study 16 28 64.3 20 22 22.9 19.8 2.05 [1.37, 2.73] 4.38
Study 17 18 69.9 15.1 23 33.6 10.2 2.83 [1.97, 3.69] 4.28
Study 18 52 68.6 16.5 31 23.7 18 2.61 [2.01, 3.20] 4.42
Study 19 44 69.3 11.7 44 28 10.3 3.71 [3.03, 4.40] 4.38
Study 20 90 97 12.4 15 6.2 17.3 6.84 [5.77, 7.92] 4.13
Heterogeneity: τ^2 = 5.10, I^2 = 97.68%, H^2 = 43.08 2.58 [1.23, 3.94]
Test of $\theta_i = \theta_j$: Q(10) = 436.75, p = 0.00
Test of θ = 0: z = 3.73, p = 0.00
Age>0
Study 2 20 62.4 19.5 40 12.7 15.3 2.92 [2.18, 3.67] 4.35
Study 4 39 52.1 17.4 31 19 16.8 1.91 [1.35, 2.47] 4.44
Study 5 54 55.5 11.5 17 31.3 21 1.68 [1.07, 2.28] 4.42
Study 8 52 76.6 15 44 11.1 11.2 4.85 [4.06, 5.64] 4.32
Study 10 38 66.3 17.4 43 16.9 16.7 2.87 [2.25, 3.49] 4.41
Study 11 50 67.1 15.4 57 24.5 16.9 2.61 [2.09, 3.12] 4.46
Study 12 30 58.2 10.4 37 12.8 14.9 3.43 [2.68, 4.18] 4.34
Study 14 18 74.5 15.6 40 11.6 12.1 4.68 [3.67, 5.69] 4.18
Study 15 33 58.5 11.9 48 33.2 9.6 2.37 [1.80, 2.94] 4.43
Study 21 36 59.4 11.6 59 11.6 15.4 3.36 [2.73, 4.00] 4.41
Study 22 36 53.4 20.4 55 25.8 17.2 1.48 [1.01, 1.95] 4.48
Study 23 32 78.9 14.5 54 22.8 14 3.92 [3.19, 4.65] 4.35
Heterogeneity: τ^2 = 1.04, I^2 = 90.93%, H^2 = 11.02 2.97 [2.36, 3.58]
Test of $\theta_i = \theta_j$: Q(11) = 108.22, p = 0.00
Test of θ = 0: z = 9.54, p = 0.00
Overall 2.79 [2.08, 3.49]
Heterogeneity: τ^2 = 2.87, I^2 = 96.30%, H^2 = 27.04
Test of $\theta_i = \theta_j$: Q(22) = 613.48, p = 0.00
Test of θ = 0: z = 7.70, p = 0.00
Test of group differences: $Q_b(1)$ = 0.26, p = 0.61
-5
0
5
10
Random-effects REML model


Treatment
Control
Study N Mean SD N Mean SD
Hedges's g with 95% CI
Weight (%)
Focal
Study 1 20 69.3 10.9 16 16.7 10.2 4.85 [3.56, 6.15] 3.96
Study 2 20 62.4 19.5 40 12.7 15.3 2.92 [2.18, 3.67] 4.35
Study 3 21 79.1 9.3 17 14.5 26 3.39 [2.40, 4.37] 4.19
Study 5 54 55.5 11.5 17 31.3 21 1.68 [1.07, 2.28] 4.42
Study 7 22 46.7 17.1 30 9.5 8.9 2.82 [2.05, 3.59] 4.33
Study 10 38 66.3 17.4 43 16.9 16.7 2.87 [2.25, 3.49] 4.41
Study 11 50 67.1 15.4 57 24.5 16.9 2.61 [2.09, 3.12] 4.46
Study 12 30 58.2 10.4 37 12.8 14.9 3.43 [2.68, 4.18] 4.34
Study 16 28 64.3 20 22 22.9 19.8 2.05 [1.37, 2.73] 4.38
Study 17 18 69.9 15.1 23 33.6 10.2 2.83 [1.97, 3.69] 4.28
Study 18 52 68.6 16.5 31 23.7 18 2.61 [2.01, 3.20] 4.42
Heterogeneity: τ^2 = 0.32, I^2 = 70.81%, H^2 = 3.43 2.81 [2.41, 3.21]
Test of $\theta_i = \theta_j$: Q(10) = 31.71, p = 0.00
Test of θ = 0: z = 13.68, p = 0.00
Generalized
Study 6 67 8 15.3 39 38.2 14.1 -2.02 [-2.49, -1.54] 4.47
Study 8 52 76.6 15 44 11.1 11.2 4.85 [4.06, 5.64] 4.32
Study 9 19 47.8 11.9 21 32.1 16.2 1.07 [0.42, 1.73] 4.39
Heterogeneity: τ^2 = 11.71, I^2 = 99.13%, H^2 = 114.63 1.29 [-2.60, 5.18]
Test of $\theta_i = \theta_j$: Q(2) = 221.88, p = 0.00
Test of θ = 0: z = 0.65, p = 0.51
Refractory
Study 4 39 52.1 17.4 31 19 16.8 1.91 [1.35, 2.47] 4.44
Study 13 49 45.5 16.2 35 34.8 17.2 0.64 [0.20, 1.08] 4.49
Study 14 18 74.5 15.6 40 11.6 12.1 4.68 [3.67, 5.69] 4.18
Study 15 33 58.5 11.9 48 33.2 9.6 2.37 [1.80, 2.94] 4.43
Study 19 44 69.3 11.7 44 28 10.3 3.71 [3.03, 4.40] 4.38
Study 20 90 97 12.4 15 6.2 17.3 6.84 [5.77, 7.92] 4.13
Study 21 36 59.4 11.6 59 11.6 15.4 3.36 [2.73, 4.00] 4.41
Study 22 36 53.4 20.4 55 25.8 17.2 1.48 [1.01, 1.95] 4.48
Study 23 32 78.9 14.5 54 22.8 14 3.92 [3.19, 4.65] 4.35
Heterogeneity: τ^2 = 3.28, I^2 = 97.05%, H^2 = 33.87 3.17 [1.97, 4.38]
Test of $\theta_i = \theta_j$: Q(8) = 204.24, p = 0.00
Test of θ = 0: z = 5.15, p = 0.00
Overall 2.79 [2.08, 3.49]
Heterogeneity: τ^2 = 2.87, I^2 = 96.30%, H^2 = 27.04
Test of $\theta_i = \theta_j$: Q(22) = 613.48, p = 0.00
Test of θ = 0: z = 7.70, p = 0.00
Test of group differences: $Q_b(2)$ = 0.92, p = 0.63
-5
0
5
10
Random-effects REML model

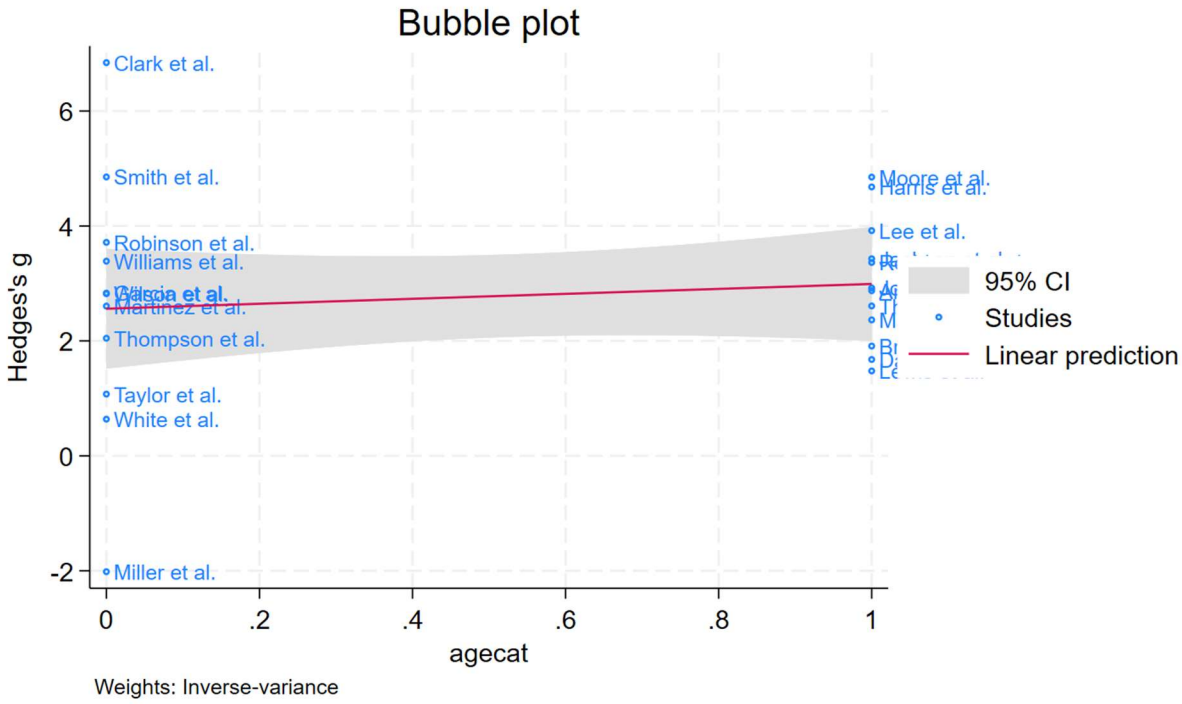
```
. meta regress ( agecat)

  Effect-size label: Hedges's g
        Effect size: _meta_es
          Std. err.: _meta_se

Random-effects meta-regression                  Number of obs   =        23
Method: REML                                    Residual heterogeneity:
                                                           tau2 =      2.95
                                                        I2 (%)  =     96.36
                                                           H2   =     27.46
                                                R-squared (%)   =      0.00
                                                Wald chi2(1)    =      0.34
                                                Prob > chi2     =    0.5572
```

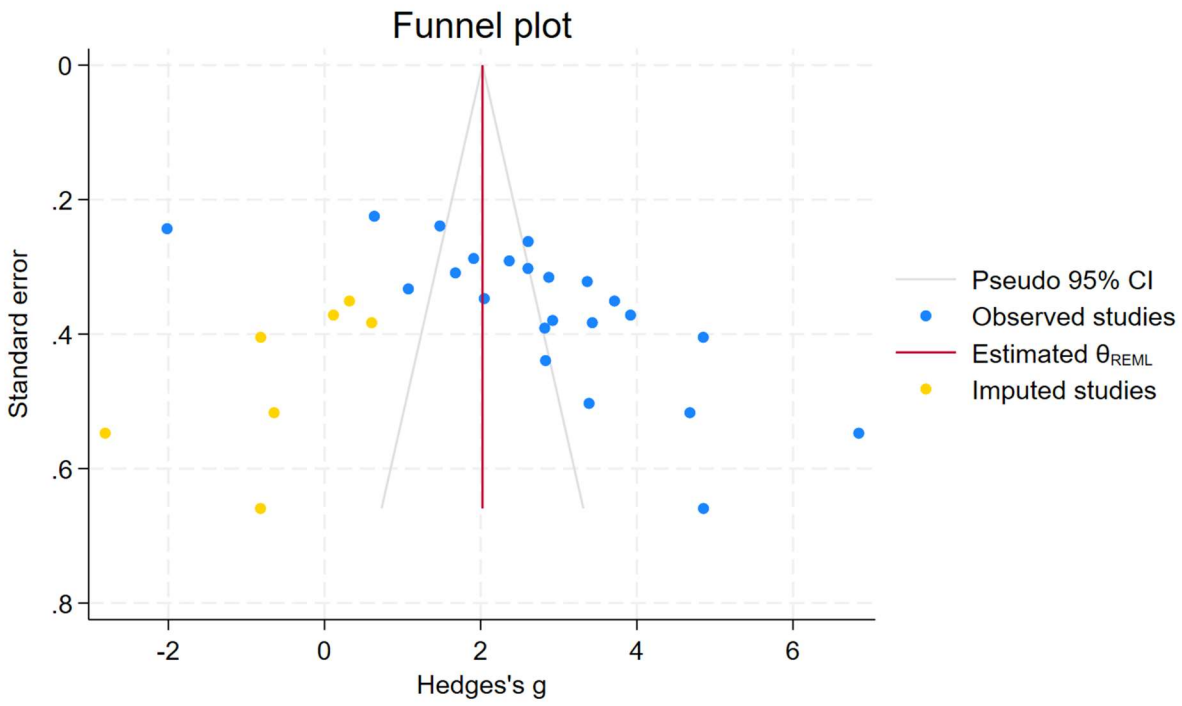
_meta_es	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
agecat	.4310387	.7343376	0.59	0.557	-1.008237	1.870314
_cons	2.560213	.5323799	4.81	0.000	1.516768	3.603659

Test of residual homogeneity: Q_res = chi2(21) = **544.97** Prob > Q_res = **0.0000**



2.

3. Assess the publication bias using the Doleman et al. method and trimfill, and interpret your findings.



The funnel plot was used to assess the symmetry of the study results and to identify possible small-study effects. Egger's test showed a statistically significant result ((z = 5.11), ($p < 0.001$)), suggesting evidence of small-study effects and possible funnel plot asymmetry.

However, this result does not necessarily confirm publication bias. Other factors, such as high heterogeneity between studies, may also explain the asymmetry. Therefore, the findings should be interpreted with caution. Since the meta-analysis included only 23 studies, the results of publication bias tests may be limited. Begg's test can provide additional evidence to support the interpretation.

. meta query

-> meta esize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

Meta-analysis setting information from meta esize

Study information

No. of studies: 23

Study label: Generic

Study size: _meta_studysize

Summary data: N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

Effect size

Type: hedgesg

Label: Hedges's g

Variable: _meta_es

Bias correction: Approximate

Precision

Std. err.: _meta_se

Std. err. adj.: None

CI: [_meta_cil, _meta_ciu]

CI level: 95%

Model and method

Model: Random effects

Method: REML

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. * 2. Funnel plot

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. meta funnelplot

Effect-size label: Hedges's g

Effect size: _meta_es

Std. err.: _meta_se

Model: Common effect

```
          Method: Inverse-variance


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. * 3. Egger's test for small-study effects

.
. meta bias, egger

 Effect-size label: Hedges's g
       Effect size: _meta_es
        Std. err.: _meta_se

Regression-based Egger test for small-study effects
Random-effects model
Method: REML

H0: beta1 = 0; no small-study effects
          beta1 =     12.41
     SE of beta1 =     2.431
               z =      5.11
     Prob > |z| =    0.0000

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.
. * 4. Begg's rank correlation test

.
. meta bias, begg

 Effect-size label: Hedges's g
       Effect size: _meta_es
```

Std. err.: _meta_se

Begg's test for small-study effects

Kendall's score = 141.00

SE of score = 37.864

z = 3.70

Prob > |z| = 0.0002